Airbus A320 Type Rating For Free
Be the best!

What is a Transponder Landing System (TLS)

Anshul Sharma
01 Jan 2022

i.meter wikipedia

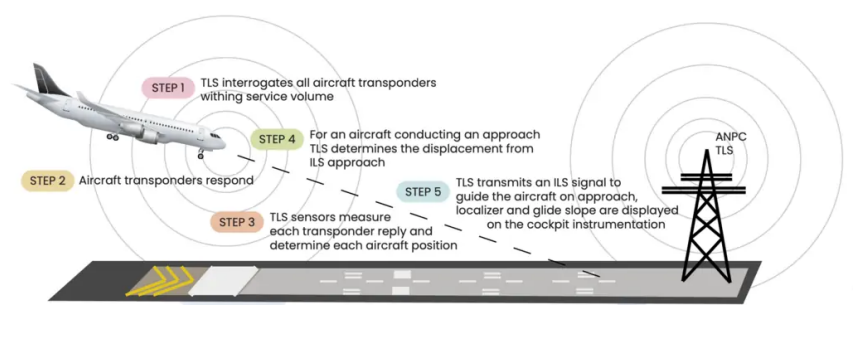
As a crew of the Flight Inspection Unit of the Airports Authority of India, our task involves undertaking commissioning, calibration, and trials of the various Navigation Aids at airports across India and in the neighboring countries. During one such exercise, we recently undertook the Transponder Landing System (TLS) trials at Safdarjung Airport, New Delhi. TLS is quite an interesting system, not many of us have read about. In the succeeding paragraphs, I have briefly explained the system.

Advanced Navigation and Positioning Corp. developed the Transponder Landing System. Though the system was conceptualized in the mid-90s, it has been under regular upgradation and development to improve further and enhance its functioning. TLS is an all-weather, precision landing system that uses existing airborne transponder and Instrument Landing System (ILS) equipment to create a precision approach at a location where an ILS would normally not be available.

A Conventional ILS system broadcasts using a number of "single purpose" antennas. One, located just off the end of the runway, provides a fan-shaped signal for azimuth direction (side to side), and another, located beside the runway, provides elevation to indicate a standard glideslope.

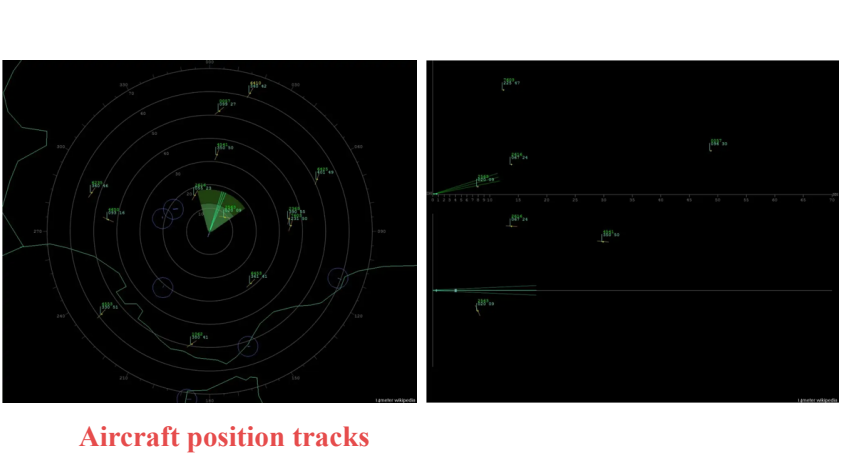
However, there are only a few components in the TLS system. The most visible are four units mounted in a 50-meter radius alongside the runway. There is a base station unit, a Calibration/Built-in-Test (BIT) unit that monitors station accuracy and integrity, and two angles of arrival antennas. The localizer and glideslope angle of arrival (AOA) sensors are used to define the flight path from the transponder system as it nears the runway. A central processor in the base station computes the aircraft's position in three dimensions, calculates where it should be in relation to the approach, and transmits corrections to the aircraft over the localizer and glideslope transmitter. The TLS complies with ICAO standards and recommended practices (SARPS) for ILS, secondary surveillance (SSR), and precision approach radar (PAR). TLS performance meets all FAA and ICAO Annex 10 requirements for a Category I approach in terms of Accuracy, Integrity, and Reliability.

The TLS detects all aircraft within the service volume by interrogating Mode-A/C /S/IFF transponders. Once the ATC clears an aircraft for the TLS approach, the pilot must tune and identify the TLS localizer frequency as they would with a traditional ILS. TLS broadcasts RF signals to the aircraft ILS equipment to provide a Category I approach. The pilot training and airborne equipment required for TLS approaches are identical to that for traditional ILS or precision approach radar. Any aircraft equipped with an ILS localizer and glide slope receiver, Horizontal Situation Indicator (HSI) or Course Deviation Indicator (CDI), and a Mode 3/A or Mode S compatible transponder can fly a TLS approach. The guidance is presented to the pilot just as it would be for an ILS approach.


STEP 1 TLS interrogates all aircraft transponders withing service volume
STEP 2 Aircraft transponders respond
STEP 3 TLS sensors measure each transponder reply and determine each aircraft position
STEP 4 For an aircraft conducting an approach TLS determines the displacement from ILS approach
STEP 5 TLS transmits an ILS signal to guide the aircraft on approach, localizer and glide slope are displayed on the cockpit instrumentation
ANPC TLS

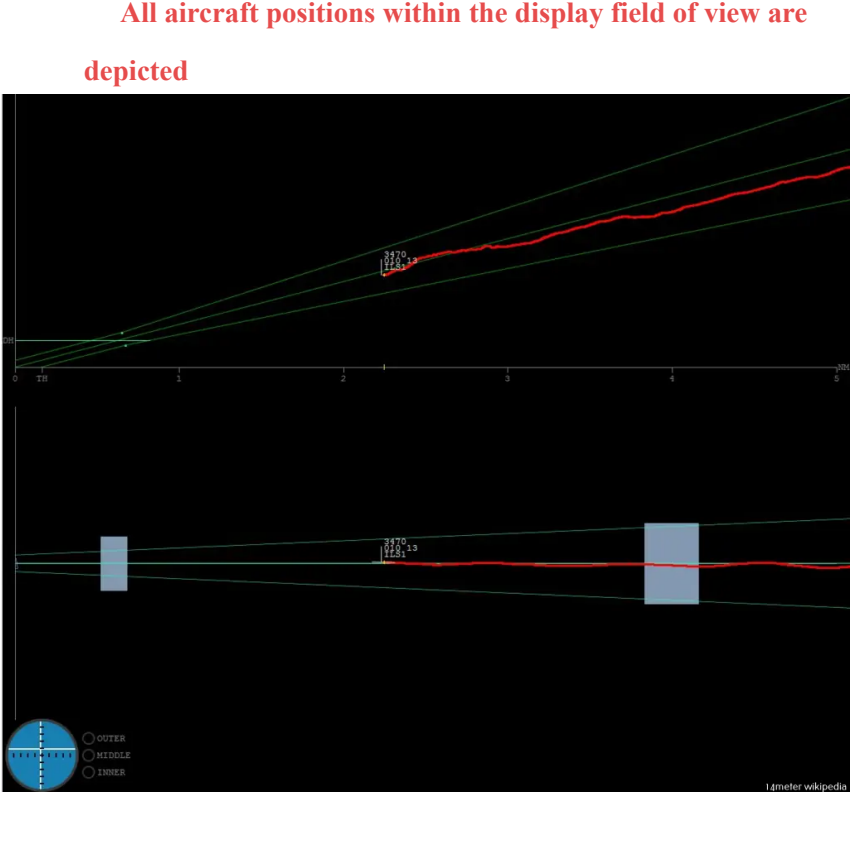
Functioning

The key to TLS technology is a compact footprint of an SSR interrogator and ground-based sensors that track an aircraft's location in range, azimuth (horizontal position), and elevation (vertical position) with very high accuracy. The range and azimuth for the surveillance tracks are computed using multilateration techniques on the arrival time of synchronous replies from the aircraft's radar beacon transponder. The elevation information for the surveillance targets uses the transponder Mode C reply. When an aircraft cleared for the TLS approach enters the approach volume, the system uses differential phase measurements on the transponder reply's carrier signal. These phase measurements are used to generate the target's precise azimuth and elevation positions. The TLS then computes the aircraft's offset from the programmed approach path and generates the ILS correction for this offset. The TLS UHF/VHF guidance transmitters then broadcast this emulated ILS signal throughout the guidance volume. The guidance appears to the pilot as needle movements on the ILS course deviation indicator in the cockpit that are identical to an ILS and can be flown by the cleared aircraft down to the minimum descent altitude. The aircraft position can also be displayed on the ground console with a format matching a Precision Approach Radar display for use in Ground Controlled Approach operations. For Ground Controlled Approach (GCA) operations, the TLS provides controllers with a secondary radar display of area aircraft traffic. For landing multiple aircraft using precision approach radar consoles, up to four individual PAR consoles can be used for controllers to guide up to four aircraft independently.



Aircraft position tracks

All aircraft positions within the display field of view are depicted



Single aircraft may be selected for each of the operator consoles to provide talk-down guidance

Key differences as Compared to an ILS

First, the approach profile for TLS is not a function of alignment between the approach path and antennas like a traditional ILS that radiates the signal without tracking the aircraft's position. The approach path is configured using virtual point technology, meaning the localizer and glideslope aiming points are mathematical points programmed into the site's TLS configuration. The TLS can inherently support offset and non-linear approach procedures where a straight-in approach is not feasible due to noise abatement or obstacle clearance issues. Virtual point technology also allows the TLS to provide a localizer signal that complies with ICAO tolerances regardless of the runway length or obstacles at the end of the runway.

Second, while a traditional ILS signal to an aircraft is impacted by multipath due to buildings and the surrounding terrain, for the TLS, the aircraft's transponder replies are affected by multipath. ANPC has developed multipath mitigation with antenna selection and signal processing. Additionally, the TLS tracking algorithms smooth the received measurements. Included in these mitigation techniques, is a site-specific calibration process that models the multipath received at each sensor array.

Third, though the TLS provides a signal identical to an ILS, it does so with equipment sitting that is more compact than an ILS. This may enable many airports to have a precision ILS approaches which would otherwise not be feasible with the traditional ILS equipment.

TLS over ILS

ADVANTAGES	DRAWBACKS
• Works over any terrain using directional antennas • The Localiser can be aligned even on short runways ending at water/obstruction. • It has multiple approach options, including offset and selectable glide slopes. • No false glide slope above 3 degrees that is possible with ILS • No false localizer capture off angle that is possible with ILS • Works with any runway length and terrain around the runway • Provides terminal area surveillance • Has PAR as a backup to recover aircraft with malfunctioning ILS receiver or no ILS receiver	• Simulates an ILS signal that is specific to one aircraft's location on a given ILS frequency • Provides guidance to only four aircraft simultaneously on discrete frequencies • Only those aircraft cleared for a TLS approach will receive proper guidance. • Other aircraft that erroneously tune to a TLS frequency without proper clearance will not receive the correct guidance. • TLS requires a human operator to acquire the aircraft at the system display console and instruct the system to provide guidance based on its transponder code

Latest Development

Trials were recently conducted for an Unmanned Aerial System (UAS) or Unmanned Aerial Vehicle (UAV) to operate and navigate in an environment devoid of a Global Navigation Satellite System (GNSS) such as the Global Positioning System (GPS). Trials were undertaken by interrogating an aviation transponder (either mode C or S) that is carried by the UAS and measuring the time elapsed for the response to multiple, ground-based antennas and using triangulation. (multilateration) to locate the transponder and, by association, the UAS. The ground-based system then routed this position information back to the UAS via the UAS's data telemetry link. The autopilot utilized this position information for navigation like it would utilize a GPS-based position report. These trials successfully demonstrated that the TLS system can guide a UAS through a series of waypoints without GPS signals. The serviced area for a TLS system may extend up to 100 Nautical miles.

The Transponder Landing System offers many advantages and can provide landing guidance at places that the traditional Instrument Landing System cannot serve. The TLS has been successfully used by helicopters to approach an Oil rig/Heliports where visual approaches in below VMC conditions have often resulted in accidents. The TLS may also effectively provide an offset and non-linear precision approach in hilly terrain where CFIT has resulted in numerous accidents.

This article was first published in the January 2022 edition of 100 Knots Magazine.

Anshul Sharma

Capt Anshul Sharma is an Ex-Indian Naval pilot with experience of flying different type of aircraft across the world. A well experienced Flying Instructor involved in training a large number of trainee pilots of Indian Navy and Indian Air Force on varied types of aircraft. He has been a DGCA approved Designated Examiner (DE) on DO 228 since 2014. Capt Anshul has had the distinction of commanding two frontline Naval Squadron in India and Mauritius. For promoting safe flying practices in Mauritius and contributing significantly to the aviation fraternity, he was honoured with the Fellowship of the Aeronautical society of Mauritius. Capt Anshul is presently employed with the Flight Inspection Unit of the Airports Authority of India.

Read next

Indigo In-Talks with Boeing to Secure 25 Wide-Body Aircraft Order

Abhishek Nayar
15 Aug 2023

©André Du-pont (Mexico Air Spotters)/wikimedia

In the ever-evolving world of aviation, exciting developments are always on the horizon. A recent wave of news has unveiled the intriguing possibility of Boeing securing an order for approximately 25 wide-body aircraft from IndiGo, India's largest airline. As of August 14, 2023, the aviation industry's attention is squarely fixed on this development, which could signify a major step forward for both Boeing and IndiGo.

Exploring IndiGo's Expanding Horizons

The Date and the Decision

On the 14th of August, 2023, aviation enthusiasts and industry observers wereon the edge of their seats as IndiGo, India's premier airline, navigates its way through a crucial decision. The stakes are high, and the implications are far-reaching as the airline charts a course for its future expansion.

The Scope of Expansion

IndiGo's ambitions know no bounds as it endeavors to broaden its horizons on the international stage. As air travel continues to evolve and global connectivity becomes more integral, airlines are vying for competitive edges. IndiGo, known for its efficient operations and customer-centric approach, is poised to embark on a journey that could redefine its role in the industry.

The Front-Runner Emerges

Boeing, a renowned name in the aviation industry, has emerged as the leading contender for a significant order of wide-body aircraft from IndiGo. According to insiders familiar with the matter, talks have been ongoing regarding IndiGo's interest in acquiring Boeing's 787 series of twin-aisle aircraft. This strategic move aligns with IndiGo's aspirations to expand its international presence, connecting more travelers to captivating destinations around the world.

A Clash of Titans: Boeing 787 vs. Airbus A330neo

As discussions unfold behind closed doors, the aircraft under consideration face an interesting standoff. Boeing's 787 series finds itself in direct competition with Airbus A330neo jets. This rivalry showcases the innovation and prowess of both aircraft manufacturers, each vying for the chance to contribute to IndiGo's expansion plans. As passengers and industry enthusiasts wait in anticipation, the decision promises to have a lasting impact on the airline's future operations.

Veiled in Secrecy: The Negotiation Process

Amidst the buzz surrounding this potential partnership, a shroud of secrecy envelops the negotiation process. Sources, preferring anonymity due to the confidential nature of the discussions, reveal that no final decision has been reached yet. This veil of secrecy adds an air of suspense to the unfolding narrative, leaving the industry to speculate about the outcomes that these negotiations might yield.

Muted Responses: Statements from Industry Giants

IndiGo, maintaining a poised stance, refrains from commenting on the swirling speculations. This silence hints at the sensitivity and significance of the ongoing negotiations. Airbus, equally enigmatic, states its policy of not engaging in commentary about potential discussions with existing or prospective clients. Boeing, too, declines to offer any insights, fostering an aura of suspense around the situation.

Unveiling the Future: What Lies Ahead?

The Implications

The outcome of these negotiations extends beyond mere aircraft acquisition. It's a reflection of IndiGo's aspirations and a blueprint for its future endeavors. The aircraft of choice will dictate the airline's capabilities, its reach, and the experiences it offers to its passengers. As Boeing and Airbus contend for this sought-after deal, their technological prowess and commitment to innovation come to the forefront.

A Glimpse into the Future

The potential collaboration between Boeing and IndiGo holds the promise of reshaping the aviation landscape. With the prowess of Boeing's 787 series and IndiGo's ambitious expansion goals, the partnership could pave the way for enhanced travel experiences and connectivity. Passengers eagerly anticipate the unveiling of the chosen aircraft, eagerly imagining the destinations they might soon explore.

Conclusion